

Topic Proposal: AI Media Localization Platform

1. Project Title

Data-Driven Adaptive Orchestration Framework for AI Video Localization: A Metric-Guided Pipeline for Automated Translation, Voice Cloning, and Lip-Sync.

2. Problem Statement & Research Gap

Currently, video localization is dominated by closed-source, commercial SaaS platforms (like ElevenLabs, HeyGen, and Dubverse). These platforms operate as "black boxes" and often force all languages through a single, static AI model. While this works for high-resource languages like English, it frequently fails for complex, low-resource Indian languages.

Proposed Contribution & Dual Base Architectures: To scientifically validate this end-to-end pipeline, this project builds upon **two highly prestigious base papers**:

- 1. For the Translation & Routing Phase:** We use the methodology from Meta AI's "**No Language Left Behind**" (NLLB) paper, which validates how to evaluate and route low-resource languages (like Indian dialects) using metrics like BLEU and chrF.
- 2. For the Audio & Lip-Sync Generation Phase:** We extend the architecture from the recent CVPR 2025 paper, "**EmoDubber: Towards High Quality and Emotion Controllable Movie**

Dubbing", which scientifically proves that end-to-end Visual Voice Cloning (V2C) is viable and can be evaluated using strict lip-sync and audio metrics.

However, EmoDubber assumes the text is already translated, and NLLB only outputs text. There is no existing framework that dynamically chooses the best model for a specific task based on actual data to bridge these two technologies.

This project's core contribution is engineering an intelligent, **Data-Driven Orchestration Framework** that connects these two phases. The system will autonomously evaluate candidate models, consult a benchmark knowledge base, and route the media to the model mathematically proven to be the best for translation (especially for low-resource Indian languages) before feeding it into the dubbing generation pipeline.

3. The Core Innovations (Why this is a strong Engineering Project)

To prove to the evaluation committee that this is a rigorous computer science project, the system introduces three novel engineering layers:

1. **Benchmark Knowledge Base:** Instead of assuming a model (like Sarvam or Whisper) is "the best", the system maintains a database of historical evaluation metrics (WER, BLEU, COMET, etc.) for various language pairs.
 2. **Adaptive Routing Engine:** When a video is uploaded, the engine detects the language, queries the Knowledge Base, and dynamically routes the audio/text to the highest-scoring model for that exact language pair.
 3. **Quality Validation & Fallback Layer:** After translation, the system evaluates the output confidence. If the output score is below a certain threshold, the system rejects it and retries the translation using the second-best fallback model.
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4. The End-to-End Workflow (Explained Simply)

1. **Language Detection & Database Query:** The system detects the video's source language and queries the Knowledge Base to identify the historically best models for Transcription (ASR), Translation (MT), and Voice Synthesis (TTS) for that language.
 2. **Transcription (ASR):** The audio is transcribed into text using the selected model (e.g., Faster-Whisper or SeamlessM4T).
 3. **Translation & Quality Validation:** The text is translated. An LLM (Large Language Model) reviews the grammar and context. If it detects hallucinations or poor grammar, it triggers a fallback to an alternate translation model.
 4. **Zero-Shot Voice Synthesis:** The validated text is converted back to speech. The system clones the original speaker's vocal characteristics so the new language sounds exactly like the original speaker.
 5. **Lip Synchronization:** The new generated audio is visually stitched back into the video, adjusting the speaker's lip movements to match the new language perfectly.
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5. Explicit Research Questions (RQs)

This project seeks to answer the following academic and engineering questions:

- **RQ1 (Model Routing):** How does dynamic, data-driven model routing improve translation and voice synthesis quality for low-resource Indian languages compared to static monolithic models?
- **RQ2 (Error Cascading):** To what extent do early-stage transcription errors propagate through the pipeline, and which module (ASR, Translation, or TTS) contributes most to final quality degradation?
- **RQ3 (Quality Assurance):** How effective is an LLM-based Quality Validation Layer in automatically catching and correcting

hallucinations or severe grammatical errors before audio synthesis?

- **RQ4 (Commercial Benchmarking):** How does this open-source orchestrated framework perform against proprietary commercial platforms (ElevenLabs, Dubverse, HeyGen) in terms of translation accuracy and lip-sync fidelity?
 - **RQ5 (SaaS Feasibility):** What are the computational constraints (GPU memory, processing latency) of running this end-to-end pipeline, and is it economically scalable for real-world deployment?
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6. How We Will Prove It Works: The Metrics (Simplified)

To eliminate assumptions, every routing decision and final output will be justified by these standard scientific metrics:

A. Transcription (ASR) Metrics

- **WER (Word Error Rate):** Measures the percentage of words the AI transcribed incorrectly. **(Lower is better).**

B. Translation Metrics

- **BLEU:** Checks how many exact words match a human translation. **(Higher is better).**
- **COMET:** A neural metric that checks if the *meaning* and context of the translation are correct, even if the exact words differ. **(Higher is better).**
- **chrF:** Evaluates at the character level, which is highly accurate for complex Indian languages where words are often glued together. **(Higher is better).**

C. Voice Synthesis (TTS) & Lip Synchronization (The EmoDubber Base)

- **SECS (Speaker Encoder Cosine Similarity):** A mathematical score proving how closely the generated AI voice matches the original speaker's real voice (as utilized in EmoDubber). **(Higher is better).**
 - **LSE-C (Lip Sync Error - Confidence):** Measures how confident the system is that the audio matches the video's lip movements. **(Higher is better).**
 - **LSE-D (Lip Sync Error - Distance):** Measures the visual distance between the generated lip shape and the ideal lip shape for that sound. Both LSE-C and LSE-D are the primary validation metrics used in the CVPR 2025 EmoDubber base paper. **(Lower is better).**
 - **MOS (Mean Opinion Score):** Human evaluation of naturalness and voice similarity.
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7. Cost, Scalability, and SaaS Feasibility

A major component of this project is evaluating if this system could actually run as a business (SaaS). We will measure:

- **Latency:** Time taken to process a 1-minute video from start to finish.
 - **Optimization:** Utilizing libraries like *CTranslate2* (which makes Whisper 4x faster and uses less RAM) to prove the pipeline can run on consumer-grade GPUs (e.g., 8GB/16GB VRAM) rather than requiring massive enterprise servers.
 - **Throughput:** Analyzing how the Quality Validation (fallback) layer impacts overall processing speed.
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8. Summary for the Evaluation Committee

This is not a project where we simply "download a model and build a UI."

It is a complex **Systems Engineering** and **Machine Learning Operations (MLOps)** project. We are building a smart routing brain that tests AI models against each other in real-time, fixes its own errors using a validation layer, tracks cascading failures, and optimizes for GPU constraints. By benchmarking against commercial giants like ElevenLabs, we are proving that an open-source, intelligently routed pipeline can achieve commercial-grade results, especially for underserved regional languages.